Problem 8.2

1000 is to be accumulated by January 1, 1995, at a compound discount of 9% per year.

(a) Find the present value on January 1, 1992.

(b) Find the value of i equivalent to d.

Solution

(a) Present value

From January 1, 1992 to January 1, 1995 is 3 years. Under compound discount at rate $d = 0.09$, the present value is

$$X = 1000(1-d)^3 = 1000(0.91)^3.$$

Thus,

$$X \approx 753.571.$$

Therefore, the present value on January 1, 1992 is

$$\boxed{\$753.57}.$$

(b) Equivalent value of i

The relationship between the effective interest rate i and the effective discount rate d is

$$d = \frac{i}{1+i}.$$

Solving for i gives

$$i = \frac{d}{1-d}.$$

Substituting $d = 0.09$,

$$i = \frac{0.09}{1-0.09} = \frac{0.09}{0.91} \approx 0.098901.$$

Hence,

$$\boxed{i \approx 0.0989 = 9.89\%}.$$